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Title:
Genome-wide signals of positive selection in human evolution

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Positive selection, background selection, human evolution.

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Abstract:

The role of positive selection in human evolution remains highly controversial. On the one hand, scans for positive selection have identified hundreds of candidate loci and the genome-wide patterns of polymorphism show signatures consistent with frequent positive selection. On the other hand, recent studies have argued that many of the candidate loci are false positives and that most genome-wide signatures of adaptation are in fact due to reduction of neutral diversity by linked deleterious mutations, known as background selection. Here we analyze human polymorphism data from the 1000 Genomes project and detect signatures of positive selection once we correct for the effects of background selection. We show that levels of neutral polymorphism are lower near amino acid substitutions, with the strongest reduction observed specifically near functionally consequential amino acid substitutions. Furthermore, amino acid substitutions are associated with signatures of recent adaptation that should not be generated by background selection, such as the presence of unusually long and frequent haplotypes (measured with *iHS* and *XPEHH*) and specific distortions in the site frequency spectrum (measured with *CLR*). We use forward simulations to argue that the observed signatures require a high rate of strongly adaptive substitutions in the vicinity of the amino acid changes. We further demonstrate that the observed signatures of positive selection correlate better with the presence of regulatory sequences, as predicted by the ENCODE Project Consortium, than with the positions of amino acid substitutions. Our results suggest that adaptation was frequent in human evolution and provide support for the hypothesis of King and Wilson that adaptive divergence is primarily driven by regulatory changes.

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Introduction.

The rate and patterns of positive selection are of fundamental interest for the study of human evolution. Population genomic studies should in principle allow us to quantify positive selection from its expected signatures in sequence polymorphism and divergence data. Surprisingly, despite sequencing of thousands of human genomes (The 1000 Genomes Project Consortium 2012) and the availability of whole genome sequences of closely related species, the extent to which adaptation has left identifiable signatures in the patterns of polymorphism in the human genome remains highly controversial (Akey 2009; Hernandez et al. 2011).

On the one hand, recent studies have identified a large number of loci showing signatures of recent selective sweeps (Voight et al. 2006; Sabeti et al. 2007; Williamson et al. 2007; Pickrell et al. 2009; Grossman et al. 2013) and McDonald-Kreitman (MK) analyses inferred that ~10-20% of amino acid changes have been adaptive in human evolution (Boyko et al. 2008; Messer and Petrov 2013). Consistently, regions of high functional density, high rate of amino acid substitutions, and low recombination all show reduced levels of neutral diversity (Cai et al. 2009; Lohmueller et al. 2011), as expected under recurrent selective sweeps in functional regions.

On the other hand, there are reasons to question that adaptation left clear signatures in the human genome. First, different scans for positive selection have identified largely non-overlapping sets of candidates (Akey 2009), which could be due to a high rate of false positives. Second, MK analyses can be confounded by a number of factors, such as perturbations left by demographic events and by the presence of slightly deleterious mutations (Eyre-Walker and Keightley 2009; Messer and Petrov 2013), and some MK analyses have failed to find evidence for adaptation in the human lineage (Eyre-Walker and Keightley 2009). Finally, it has been shown that background selection (BGS) (Charlesworth et al. 1993), a process in which deleterious mutations remove linked neutral variation from the population, reduces levels of polymorphism in regions of higher functional density and low recombination, providing an alternative explanation for the observation of these correlations in the human genome.

One signature of positive selection - lower levels of neutral variation near functional substitutions (Andolfatto 2007; Macpherson et al. 2007; Cai et al. 2009) - is not generally expected under BGS and should therefore provide the clearest genomic evidence for the action of positive selection. While this signature was found in the human

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genome by Cai et al (Cai et al. 2009), it could not be detected by two recent studies using the newest large-scale datasets of human diversity (Hernandez et al. 2011; Lohmueller et al. 2011). In particular, Hernandez *et al.* (Hernandez et al. 2011) searched for lower levels of neutral diversity near functional substitutions by contrasting levels of neutral diversity near nonsynonymous compared to synonymous substitutions. They did not find this signature in the human genome, and, moreover, found that diversity might in fact be marginally higher near nonsynonymous substitutions. Simulations showed that this puts sharp limits on the amount of adaptation by classic selective sweeps in recent human evolution (Hernandez et al. 2011).

However, it is likely that the study design of Hernandez *et al.* (Hernandez et al. 2011) (also implemented by Sattah *et al.* (Sattath et al. 2011) in Drosophila) is strongly biased against finding signatures of positive selection in the human genome and all other genomes with sharply variable levels of genomic constraint. This is because, as we show in the Results, nonsynonymous substitutions in the human genome tend to be located in regions of weaker constraint and thus weaker BGS compared to synonymous substitutions. These differences in levels of BGS should elevate neutral diversity near nonsynonymous compared to synonymous substitutions. The approach of Hernandez *et al.* would thus detect positive selection only if the reduction of diversity due to positive selection near nonsynonymous substitutions happens to be greater than the initial difference in the opposite direction due to BGS.

Here we employ a number of more sensitive approaches in the search for signatures of positive selection while attempting to reduce the confounding effects of BGS to the greatest extent possible. Our results suggest that positive selection was frequent in human history and might have involved adaptive mutations of substantial selective effect. We estimate that on the order of a few hundred of strong adaptive events are likely to be detectable in the human genome, consistent with the latest scan for positive selection (Grossman et al. 2013). Moreover, we provide evidence that the majority of adaptive substitutions were due to cis-regulatory rather than protein-coding changes, consistent with the King and Wilson (King and Wilson 1975) hypothesis that adaptive divergence is primarily driven by regulatory changes.

Results.

The search for signals of positive selection in the human genome is complicated by highly variable levels of functional constraint, and thus BGS, along the genome. To be

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144 able to detect positive selection, the reduction of neutral polymorphism near
145 nonsynonymous substitutions needs to be stronger than the elevation of polymorphism
146 due to weaker BGS in the same regions. This bias against detecting evidence of positive
147 selection should be particularly strong when levels of neutral polymorphism near
148 synonymous and nonsynonymous substitutions are contrasted, as in the approach
149 pioneered by Hernandez *et al*. (Hernandez et al. 2011) and Sattath *et al.* (Sattath et al.
150 2011). First, these approaches have limited power in the human genome as the majority
151 (~65%) of all synonymous substitutions are located extremely close (less than 0.02 cM)
152 to nonsynonymous ones (Methods). What is more troubling is that the synonymous
153 substitutions that are located far from nonsynonymous substitutions mark regions of
154 particularly strong selective constraint and thus particularly strong BGS. Specifically, we
155 find that 82% of synonymous substitutions are found within conserved segments of the
156 genome predicted by phastCons (Siepel et al. 2005) (Methods), versus only 56% of
157 nonsynonymous ones.
158 The reason for this difference between synonymous and nonsynonymous
159 substitutions is that selectively constrained regions, by definition, lack nonsynonymous
160 substitutions, but still allow changes at less constrained synonymous sites. Because
161 constrained regions have stronger BGS, this means that BGS could in principle reduce
162 diversity more drastically near synonymous than near nonsynonymous substitutions to
163 an extent sufficient to mask positive selection. To test whether this could happen in the
164 human genome, we simulate BGS in regions with varying levels of constraint under
165 currently accepted population size, recombination and selection parameters
166 (Supplemental Material). The simulated regions contain 10% of potentially functional
167 sites, and we vary the proportion of functional sites that are under constraint from zero to
168 100%. Note that even in simulations with 100% functional sites under constraint, we use
169 a Distribution of Fitness Effects (DFE) with many mutations being virtually neutral,
170 making our analysis overall conservative (Supplemental Material). These simulations
171 show that diversity in the strongly constrained regions can be reduced by as much as
172 10% on average (Supplemental Fig. 1). This decrease is similar or even greater than
173 what is expected to be generated by positive selection (Hernandez et al. 2011), and thus
174 could have been sufficient to hamper previous attempts to detect positive selection near
175 nonsynonymous changes. We therefore believe that using the synonymous substitutions
176 as the control for detecting reduction of neutral polymorphism near nonsynonymous

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177 substitutions in the functionally heterogeneous human genome might be unduly
178 conservative.
179 Below we devise sensitive methods for the detection of positive selection in the
180 human genome. We first search for reduction of neutral polymorphism near all and near
181 functionally important nonsynonymous substitutions. We match regions near and far
182 from nonsynonymous substitutions by levels of BGS measured using a variety of
183 correlates of BGS, such as levels of functional constraint and recombination. We focus
184 specifically on regions of low BGS, because in these regions the bias against finding
185 positive selection should be the weakest. In addition to level of diversity, we also employ
186 haplotype-based statistics *iHS* and *XPEHH*. Unlike the overall level of neutral
187 polymorphism employed in the first set of tests, we demonstrate that these haplotype
188 statistics are virtually insensitive to BGS and, as a result, that their extreme deviations
189 are predictive of recent and strong selective sweeps.
190 All data analyses are carried out with the 1000 Genomes phase 1 data 20100804
191 release (http://www.1000genomes.org). Levels of neutral diversity are calculated as the
192 average pairwise heterozygosity at putatively neutral sites, scaled by divergence
193 between human and macaque. Human specific substitutions at synonymous and
194 nonsynonymous sites are inferred from human-chimpanzee-orangutan alignments
195 (Methods).
196
197 *Choosing analysis windows*
198 BGS is expected to be stronger in regions of low recombination: consistent with this, the
199 correlation between neutral diversity and recombination rate measured in 500 kb
200 windows sliding every 1000 kb is strong and positive (n=2,247, Spearman's ρ=0.45,
201 $P<2\times10^{-16}$). Note that all correlations reported in this section use 500 kb windows that
202 are at least 500 kb from each other to ensure independency of the estimates to the
203 greatest extent possible. BGS should also be stronger in regions of high functional
204 constraint. We can measure functional constraint using multiple variables, all of which
205 show strong negative correlations with levels of neutral diversity in 500 kb windows,
206 including (i) density of coding sequences (CDS) (n= 2,247, ρ=-0.22, $P<2\times10^{-16}$), (ii)
207 density of conserved coding sequences (CCDS) and non-coding sequences in all
208 mammals or just in primates according to phastCons (ρ=-0.22, $P<2\times10^{-16}$ and ρ=-0.25,
209 $P<2\times10^{-16}$, respectively), and (iii) the density of UTRs (ρ=-0.23, $P<2\times10^{-16}$). All of these
210 correlations were computed as partial correlations controlling for recombination rate. In

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addition, we also find a strong negative partial correlation between diversity and GC content controlling for recombination (ρ=-0.2, $P<2x10^{-16}$) that might be related to high GC content of coding regions (Lander et al. 2001) or some other property correlated with the GC content.

The segments of conserved DNA identified by phastCons are shared by mammals and/or primates and represent averaged constraint over long evolutionary periods of time. The density of mammalian and primate conserved sequences being equal, large regions that are particularly devoid of human-specific nonsynonymous substitutions may still be under stronger constraint. To detect such regions of unusually strong constraint, and thus BGS, we plot the distribution of distances to the nearest amino acid substitution in the human genome in all regions that have CCDS density greater than 0.1% (to make sure there are CCDS in the windows). 67.4% of the windows are located less than 0.1 cM away from a human-specific amino acid substitution, 30% are located 0.1 to 1 cM, and 2.6% are located farther than 1 cM. These latter windows may represent regions of unusually strong constraint.

We quantify whether the regions of moderate to high functional density (CCDS density>0.5%) located far (> 1 cM) from any amino acid substitution are indeed subject to stronger BGS by conducting a bootstrap procedure (Methods). For each window located between 0.1 and 1 cM away from an amino acid change we match a randomly sampled window located 1 cM or farther whose functional density, GC content, and recombination do not differ by more than empirically fixed thresholds compared to the 0.1cM - 1 cM window. Windows less than 0.1 cM away are excluded from this comparison since they are the ones most likely to be affected by positive selection and we want to focus only on BGS as a function of distance to the nearest amino acid change. Thresholds of the bootstrap are adjusted such that 0.1 cM to 1 cM and >1 cM windows have similar average functional density, GC content and recombination rates. Windows for which no good match can be found are excluded (the detailed bootstrap procedure is described in the Methods).

Neutral diversity is indeed substantially reduced in regions that are located more than 1 cM away from any amino acid substitution, controlling for functional density, GC content, and recombination (Methods). Overall, the reduction is 7% (randomization test, $P=8.6x10^{-3}$) and becomes even stronger (~15%; randomization test, $P=2.4x10^{-2}$) in regions where recombination rates do not exceed 1 cM/Mb. This is consistent with our interpretation that regions of substantial functional density that are located very far from

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any amino acid change are more constrained in a way that cannot be accounted for using average conservation in more distant mammalian and primate species. Below we exclude the 2.6% of the windows that are located farther than 1 cM from an amino acid substitution unless stated otherwise.

The near-vs-far test.

One key expectation under positive selection is a reduction of neutral diversity near functional substitutions. We first test this prediction by contrasting neutral diversity in 500kb windows near (< 0.1 cM) compared to far (> 0.5 cM) from any of the 21,278 amino-acid substitutions we identified in the human lineage (Methods). Importantly, the windows are matched by all parameters associated with BGS that we described above.

We first carry out this test in the regions with low density of conserved coding sequences (CCDS density<0.5%) and thus weak effects of BGS. This analysis reveals a substantial 5% decrease of neutral diversity near amino acid changes (Fig. 1A; randomization test $P = 6x10^{-3}$; Methods). As expected under frequent positive selection, the decrease is more pronounced in low recombination regions (<1 cM/Mb), where the decrease of diversity is 8% on average ($P=1.5x10^{-2}$). The decrease is stronger in the Asian (9.5%, $P=1.2x10^{-2}$) and European (9.5%, $P=1.2x10^{-2}$) populations than in Africa (5%, $P=7.5x10^{-2}$) (Fig. 1B, C, D).

When we include regions of higher conserved coding density (>0.5%), as well as those located more than 1cM away from any amino acid substitution, we fail to detect any decrease in neutral diversity near amino acid substitutions. In fact, we find the opposite pattern of, on average, 4% higher diversity near amino acid substitutions ($P=4.6x10^{-2}$), reminiscent of the results of Hernandez *et al.* (Hernandez et al. 2011). This suggests that BGS can indeed obscure signatures of positive selection in the human genome, making it essential to control for BGS and reduce its effects as much as possible when searching for positive selection.

The functional-vs-nonfunctional test.

The regions of low BGS in the near-vs-far test above correspond to ~30% of all the regions in which the test can be applied in principle, and ~17% of the genome in total (290 Mb of “near” and 236 Mb of “far” windows; supplemental Table 1). We are thus unable to apply this test to the majority of the genome. In addition, the choice of the threshold of CCDS < 0.5% is somewhat *ad hoc* and was driven by the need to have

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enough windows for the bootstrap procedure while reducing the effect of BGS as much as possible.

In order to find additional signatures of positive selection that are less sensitive to BGS and can be applied to more of the genome, we modify the near-vs-far test to compare windows that have the same overall number of amino acid substitutions (i.e. all the windows are "near"), and then contrast the windows that differ by the presence or absence of predicted *functionally consequential* substitutions, as defined by PolyPhen-2 (Adzhubei et al. 2010) (Methods). We reason that predicted functionally consequential substitutions are more likely to be adaptive than predicted neutral ones and should be associated with a more pronounced reduction of neutral polymorphism in their vicinity. At the same time, controlling for the total overall number of nonsynonymous substitutions in a window naturally controls for the variation in BGS.

We compare neutral diversity in 500kb windows either near predicted functional amino acid substitutions (<0.1 cM) or near predicted neutral amino acid substitutions (<0.1 cM from a neutral substitution and >0.5 cM from a functional one). The matching windows must have the same (plus/minus one) total number of amino acid substitutions. In addition, we again control for the key genomic variables (densities of coding, conserved coding and non-coding sequences, recombination rate, and GC content; Methods). In total, this functional-vs-nonfunctional test includes 823 Mb near functional substitutions and 768 Mb near nonfunctional ones (~50% of the genome in total; supplemental Table 1) and therefore greatly extends the span of the human genome we are able to analyze.

Using this test we find that neutral diversity is decreased by ~3% on average near functional compared to nonfunctional amino acid substitutions (Fig. 2A). This decrease is statistically significant but marginally so (randomization test $P=4.8\times10^{-2}$). As expected, the decrease of diversity is more pronounced in regions with low rates of recombination (5% on average, $P=2\times10^{-2}$; <1 cM/Mb) (Fig. 2A). The decrease is again weaker in the African population (3%, $P=0.1$) compared to Asian (5%, $P=4.5\times10^{-2}$) and European populations (7%, $P=5\times10^{-3}$) (Fig. 2B, C, D). Together with the near vs. far test this test further suggests a detectable effect of recent positive selection on human genetic diversity.

The omnibus near-vs-far and functional-vs-non-functional test.

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The near-vs-far test (Fig. 1) and the functional-vs-nonfunctional test (Fig. 2) search for different signals in the data and should be independent of one another. Indeed, all regions in the functional-vs-nonfunctional test are located *near* an amino acid substitution, and thus they are all in the "near" category in the near-vs-far test. The fact that the "near" regions have lower diversity than the "far" regions should not affect the results of the test that looks only within the "near" regions. In addition, we confirm by simulation that the finite number of regions used in the bootstrap procedure does not generate spurious correlations between the two tests (Supplemental Material).

The independence of the two tests allows us to combine them into a single, omnibus test and calculate a joint *P*-value (Fig. 3). In all human populations the observed combined decreases are highly statistically significant, as shown by the *P*-values of the combined randomization test in Fig. 3 (all populations combined $P=3\times10^{-4}$; Asian, $P=2\times10^{-4}$; African, $P=7\times10^{-3}$; European $P=2\times10^{-4}$). Even in Africa where the signal of positive selection is consistently weaker in both the near-vs-far and the functional-vs-nonfunctional tests, the probability of both observed decreases by chance is less than 1%. In European and Asian populations the same probability is lower than 0.1%. Taken together, these results strongly suggest that positive selection has significantly decreased neutral diversity in the human genome.

Extreme values of XPEHH and iHS near and far from nonsynonymous substitutions. Both positive selection and BGS are expected to reduce the overall level of neutral polymorphism. Yet only positive selection, and not BGS, is expected to drive individual haplotypes to unusually high frequencies. Tests based on the presence of unusually frequent and long haplotypes, such as *iHS* (Sabeti et al. 2002; Voight et al. 2006) and *XPEHH* (Sabeti et al. 2007), should therefore be insensitive to BGS and provide a less confounded approach for the systematic detection of positive selection in the genome.

We first use extensive forward simulations to confirm this intuition. We use SLiM (Messer 2013) to simulate 4 Mb regions that include a 100 kb central region where deleterious mutations occur with a predefined strength of selection and rate (Supplemental Material). We analyze a range of distributions of selective effects of deleterious mutations (Supplemental Fig. 2), including a gamma distribution that matches our best current estimate of the distribution of fitness effects of functional mutations in the human genome (Keightley and Eyre-Walker 2007). As expected, BGS has a strong effect on levels of diversity (Fig. 4), but has no detectable effect on *XPEHH*

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and only marginal effect on *iHS*. BGS slightly decreases the variance of *iHS* values, thereby making scans for positive selection using extreme values of *iHS* conservative in the presence of BGS.

We modify the near-vs-far test by using extreme values of *iHS* and *XPEHH* instead of overall levels of neutral diversity near and far from amino acid substitutions as a measure of positive selection. For *iHS*, we consider the distribution of absolute values to capture adaptation driven by both ancestral and derived alleles and to avoid issues due to potential mispolarization of ancestral states. Specifically, we compare the most extreme values of *XPEHH* and *iHS* near and far from amino acid substitutions (Fig. 5). For example, we use the 10% of windows with the highest *iHS* values near amino acid changes and use the average *iHS* for these windows. We then compare this value with the average *iHS* of the 10% most extreme windows far from amino acid changes (Fig. 5). This comparison is repeated using either the 5%, 2% or 1% most extreme windows (Fig. 5). We use values of *iHS* and *XPEHH* calculated for the HGDP panel by Pickrell *et al*. for Bantu, East Asia and Europe populations (Pickrell et al. 2009). As before, the "near" windows are less than 0.1 cM and the "far" windows are more than 0.5 cM from any amino acid substitution. We control for levels of recombination and coding density in the bootstrap procedure. The significance of the differences between near and far windows is again calculated using the randomization test (Methods).

Fig. 5 shows clear signatures of positive selection in the *iHS* and *XPEHH* modification of the near-vs-far test. *iHS* shows significantly more extreme values near amino acid changes in all three tested populations (Fig. 5, upper row). In line with our prediction this pattern is more pronounced in low recombination regions (<0.5 cM/Mbp) (Fig. 5, right side of histograms), especially in the African population. In order to increase the statistical power, we also compare the maximum values of *iHS* in East Asians and Europeans in each window near and far from amino acid changes and find that this test yields even more significant results.

Results are essentially the same when using the *XPEHH* modification of the near-vs-far test (Fig. 5, second row). We choose the African population as the reference population and apply the *XPEHH* modification of the near-vs-far test in the two remaining populations. The results are significant in both populations and again become more pronounced in the low recombination regions (<0.5 cM/Mbp) and when the maximum value of XPEHH in the two populations is used as a test statistic. Similar results are also

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379 obtained using the *CLR* test (Williamson et al. 2007) (Supplemental Material,
380 Supplemental Fig. 3).
381 Because *iHS* and *XPEHH* are insensitive to BGS we were able to carry out these
382 tests even in regions that have high coding density and in which the tests that rely on the
383 overall level of polymorphism are too biased by BGS against the detection of positive
384 selection. Specifically, the "near" windows in the *iHS* and *XPEHH* tests represent a total
385 of 1.56 Gb and "far" windows represent a total of 618 Mb, extending the analysis to
386 ~70% of the human genome (Supplemental Table 1).
387
388 *Forward simulations of positive selection.*
389 We run forward simulations of positive selection using SLiM (Messer 2013) (Methods) in
390 order to determine the frequency and strength of selective sweeps required to decrease
391 neutral diversity near amino acid substitutions between 2% and 9.5% in 500 kb windows,
392 as observed in the data (Figs. 1 and 2). In particular, we focus on regions of low
393 recombination (<1 cM/Mb) and simulate adaptation with three different rates of adaptive
394 amino acid substitutions (proportion of substitutions that are adaptive α=10%, 20% and
395 40%) and two different selection regimes (selection coefficient s = 0.01 and 0.05).
396 Surprisingly, the amount of strong positive selection required to explain the observed
397 reduction in diversity is very high (Fig. 6). The observed 9.5% reduction in Europe and
398 Asia is similar to the average reduction expected if 40% of amino acid substitutions were
399 adaptive with a selection coefficient of s=0.05; they are in the high range if 10 or 20% of
400 amino acid changes were adaptive with s=0.01.
401 This rate appears higher than that estimated with MK approaches, which predict
402 that 10-20% or fewer of amino acid changes (Boyko et al. 2008; Messer and Petrov
403 2013) were adaptive in the human lineage. While MK estimate include both strongly
404 (s>0.01) and weakly (s<0.001) selected substitutions, our simulations suggest that at
405 least 10% of strongly selected (s=0.01) amino acid substitutions would be required to
406 obtain the observed decrease in diversity. This implies that either at least half of the
407 adaptive amino acid substitutions were driven by strong selection, or, alternatively, that
408 the majority of adaptive changes are not amino acid substitutions themselves, but
409 instead are adaptations at nearby, possibly regulatory, sites.
410
411 *Adaptation is centered at the ENCODE-defined regulatory elements*

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412 The above simulations suggest that adaptation by amino acid substitutions is unlikely to
413 generate all of the observed signatures of adaptation. We therefore search for
414 adaptation at regulatory regions by focusing on the regulatory elements defined by the
415 ENCODE Project Consortium (ENCODE-defined regulatory elements, or ERE) (Gerstein
416 et al. 2012). ERE density in our analysis is the density of elements predicted as DNASE
417 I hypersensitive sites and also as transcription factor binding sites identified via ChIP-
418 seq by the ENCODE Project Consortium (Gerstein et al. 2012). Note that our strict
419 definition of ERE elements leaves us with only 4% of the positions in the analyzed
420 windows. Moreover the ERE content correlates strongly with coding density (n=2,189,
421 Spearman's ρ=0.73, $P<2\text{x}10^{-16}$), suggesting that EREs are indeed often functional.
422 We examine the correlation between the density of ERE and *iHS* in three
423 populations of the 1000 Genomes phase 1 project (Supplemental Material). In Europe
424 and Asia, absolute values of *iHS* correlate positively with ERE density (Fig. 7B,C), but
425 the correlation is more subtle in Africa, where it becomes positive only in low
426 recombination regions (Fig. 7A). The correlation is notably stronger in regions with low
427 recombination rates, as expected under frequent positive selection.
428 Because ERE density correlates with the coding density and the coding density
429 correlates with *iHS* (Fig. 7), it is important to disentangle the respective contributions of
430 coding and regulatory sequences to the observed signals of recent positive selection. In
431 order to do so we calculate the reciprocal partial correlations between (i) *iHS* and ERE
432 density controlling for coding density and (ii) *iHS* and coding density controlling for ERE
433 density. When using the whole genome regardless of recombination, partial correlations
434 between *iHS* and ERE or coding density are weak and inconsistent between different
435 human populations, being either positive in Asia or negative in Africa (Fig. 7A,B,C). In
436 low recombination regions (<0.5 cM/Mb), where the effects are expected to be the
437 strongest and clearest, the results are striking (Fig. 7A,B,C; Supplemental Table 2):
438 while the partial correlation between *iHS* and ERE density appears virtually independent
439 of coding density, the correlation between *iHS* and coding density disappears entirely
440 once controlling for ERE density. These results are robust to the amount of overlap
441 between the windows used to measure the correlations (Supplemental Table 2). In
442 particular, we measure significant correlations with windows that are located at least 1
443 Mbp from each other and thus expected to provide largely independent values of all
444 statistics (Supplemental Table 2). These results provide evidence that many signals of

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445 positive selection in the human genome may indeed be due to adaptation centered in
446 regulatory rather than in coding sequences.
447
448 **Discussion.**
449 In this study, we have used a number of independent approaches to search for
450 signatures of positive selection in the patterns of variation in the human genome. Our
451 results show that BGS can inhibit the detection of even frequent positive selection in
452 humans. The key prediction of recurrent positive selection is that neutral polymorphism
453 should be lower in regions with more functional, for instance, nonsynonymous
454 substitutions. Perhaps counterintuitively, BGS is expected to generate the opposite
455 signature: regions of the genome with coding sequences but very few nonsynonymous
456 substitutions are likely to exhibit stronger BGS and lower levels of neutral polymorphism.
457 This means that the standard approaches that search for adaptation using the signature
458 of low levels of polymorphism next to nonsynonymous substitutions, such as those of
459 Andolfatto *et al*., Macpherson *et al*., Cai *et al*., Hernandez *et al.* and Sattath *et al*.
460 (Andolfatto 2007; Macpherson et al. 2007; Cai et al. 2009; Sattath et al. 2011), are likely
461 to underestimate the effect of positive selection. This underestimation is likely to be
462 marginal in small and functionally dense genomes, such as that of Drosophila, where
463 levels of BGS are expected to be homogeneous along the genome. However, in larger
464 genomes with heterogeneous distribution of functional sequences, such as that of
465 humans, the levels of BGS vary sharply along the genome and this bias against finding
466 signatures of positive selection can become profound. We confirmed this assertion using
467 forward simulations of BGS in the human genome.
468 We were able to detect lower levels of polymorphism near nonsynonymous
469 substitutions. Surprisingly, our results indicate that carefully matching windows near and
470 far from nonsynonymous substitutions for a number of factors known to correlate with
471 selective constraint and diversity in general is essential but not sufficient to fully control
472 for BGS. Indeed, decreased diversity near nonsynonymous substitutions is apparent
473 only when genomic regions of low functional density and hence weak BGS are analyzed.
474 The opposite pattern of higher diversity near nonsynonymous substitutions is seen in
475 regions of high functional density and hence strong BGS. Our interpretation of this
476 pattern is that the very fact of observing a nonsynonymous substitution carries
477 substantial information not only about rates of adaptation but also about the lower level
478 of constraint and thus BGS. The latter effect becomes dominant in regions with strong

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BGS in general. Note, that although on balance we believe this is the most parsimonious explanation of the data and is consistent with the results based on haplotype statistics that are insensitive to BGS, we cannot exclude that some unknown variable correlating with average pairwise diversity might affect our results or that windows in regions of low functional density exhibit drastically different distributions of fitness effects of deleterious mutations.

Although unlikely, it is therefore still in principle possible to imagine scenarios where BGS alone could explain our results with average pairwise diversity. Thus it is essential that we were able to detect positive selection using the presence of long and frequent haplotypes that are unlikely to be mimicked by BGS. Indeed, we conducted extensive simulations of BGS under varying rates and patterns of deleterious mutation and showed that tests of selection based on the presence of long and frequent haplotypes (*iHS* and *XPEHH*) are insensitive to BGS. As expected under positive selection, we detected significantly more extreme values of *iHS* and *XPEHH* near amino acid substitutions. Because these statistics are insensitive to BGS we were able carry out this analysis systematically on a genome-wide scale, without beeing restricted to only the regions with low functional density, as in the case of the near versus far test using average pairwise diversity.

All the evidence together suggests that positive selection left detectable effects on patterns of variation in the human genome. However, it is also clear that these patterns are challenging to detect and quantify due to a number of factors in addition to BGS. First, demographic perturbations such as bottlenecks and admixture can generate variability in levels of polymorphism and haplotype structure. Yet it is hard to imagine a scenario in which these demographic perturbations would affect windows near amino acid substitutions differently from those that are far from amino acid substitutions in the long history of evolution since divergence of humans and chimpanzees. Indeed, the vast majority of the amino acid substitutions happened long ago, prior to any demographic event in question. In addition, the windows near and far from amino acid substitutions that are used in the comparisons have had exactly the same demographic history. Thus the main effect of demography is to increase variance in levels of polymorphism both in windows near and far from amino acid substitutions, but it is unlikely to generate false positives by itself. Both the forward simulations and permutation tests carried out here highlight substantial variance in levels of polymorphism due to drift, which strongly limits our ability to obtain precise estimates of the rate and strength of recent positive

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513 selection. Second, it is very important to use windows with a large physical size (500 kb
514 in our case) to correctly control for BGS (Methods). This means however that our
515 analysis is likely limited to detecting only the effects of strong recent positive selection,
516 given that only selected mutations with selection coefficients on the order of 1% or larger
517 can affect diversity in regions of several hundreds of kilobases (Sabeti et al. 2006). This
518 bias against weaker positive selection further complicates attempts to precisely quantify
519 the rate and strength of positive selection. Therefore our results suggest that BGS may
520 make it far more difficult to distinguish and quantify the rates of weakly and strongly
521 advantageous mutations, as could be done in Drosophila (Macpherson et al. 2007;
522 Sattath et al. 2011). Third, patterns of recombination may also make it more difficult to
523 detect positive selection in human compared to Drosophila (Myers et al. 2005). Indeed,
524 recombination rates are known to be more heterogeneous in human than in Drosophila,
525 and this may add even more variance in our analyses.
526 In our simulations of positive selection we estimate that the observed decrease of
527 diversity of approximately 10% near amino acid substitutions would correspond to
528 roughly a hundred strong sweeps (s=0.05) in the past 100,000 years. However,
529 confidence intervals in our simulations (Fig. 6) and the lack of knowledge of the
530 distribution of fitness effects of advantageous mutations (Fig. 6) make it very hard to
531 estimate the rate of adaptation precisely. While our results do exclude the possibility that
532 adaptation had no effect on neutral diversity, they provide only a very rough order of
533 magnitude estimate for the rate of recent positive selection.
534 It is also worth noting that although signals of positive selection are detectable in
535 all tested populations, these signals are systematically stronger in the out-of-Africa
536 populations. This pattern will require further investigation, as it could be due to many
537 different and non mutually exclusive reasons such as variation in demography,
538 differences in patterns of linkage disequilibrium, different rates of adaptation, different
539 patterns of monogenic and polygenic adaptation in different populations, and differences
540 in the proportion of adaptation from *de novo* mutations versus from standing genetic
541 variation.
542 In this study we detect and quantify a number of signatures left by apparently
543 abundant recent positive selection in the genome-wide signals in diversity in humans.
544 We also provide suggestive evidence that positive selection may have been driven
545 largely by regulatory rather than coding sequences. The primary evidence for this comes
546 from two observations. First, haplotype signatures of positive selection captured by

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extreme values of *iHS* correlate better with the density of ENCODE regulatory elements (ERE) than with the density of coding sequences (Fig. 7). Second, admittedly simplified simulations suggest that without a substantial number of strongly advantageous regulatory substitutions taking place in the same regions as the amino acid substitutions, the total impact of positive selection on diversity would require a seemingly unreasonable number of strongly advantageous amino acid substitutions. Because windows near amino acid substitutions have 21% more human-specific fixed regulatory (ERE) substitutions compared to the windows far away from amino acid substitutions (as defined in the near-vs-far test, Fig. 1) and because ERE substitutions are approximately 30 times more common than amino acid substitutions, even a modest difference in the rate of adaptation within EREs between near and far windows could potentially explain our results for the near vs. far test.

Although our results support that most adaptation is regulatory, the possibility also remains that rates of coding adaptation obtained by MacDonald-Kreitman approaches may be strong underestimates, as previously discussed by Eyre-Walker and Keightley (Eyre-Walker and Keightley 2009). This said, our conjecture is consistent with recent results showing that adaptation between different human populations may have been driven primarily by regulatory rather than coding differences as well (Fraser 2013). The challenge for the future is to better quantify and identify the nature of recent human-specific adaptations. This will likely require improved modeling of BGS in the human genome, based on a deeper knowledge of the distribution of fitness effects and its variability along the human genome.

Methods

Human-specific nonsynonymous and synonymous fixed substitutions

Human-specific nonsynonymous and synonymous substitutions were obtained using human-chimpanzee-orangutan coding DNA sequence (CDS) alignments. Human CDS are first extracted from the Ensembl v64 database (Flicek et al. 2012) (http://www.ensembl.org/). For each gene, only the longest CDS is retained. Human longest CDS are then mapped onto the chimpanzee and orangutan genomes using BLAT (Kent 2002) (protein-protein BLAT, 60% minimum identity). The best, highest identity chimpanzee and orangutan BLAT hit sequences are then mapped back on the human genome. Only those human-chimpanzee and human-orangutan best reciprocal hits are retained for further analysis. Extracting chimpanzee and orangutan CDS from

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their respective genomes using BLAT instead of directly using Ensembl annotations ensures that the sequences used during subsequent global alignment steps have good local similarity. The analysis is further restricted to those best BLAT reciprocal hits that coincide with Ensembl v64 one-to-one orthologs. A total of 17,237 CDS multiple alignments are finally obtained using Prank (Loytynoja and Goldman 2008) under the codon evolution model settings. Prank used with its codon evolution model was previously shown to be the most accurate solution to align CDS (Fletcher and Yang 2010). From these alignments, a total of 27,538 and 40,709 nonsynonymous and synonymous human-specific substitutions are identified, respectively. This includes only those cases where chimpanzee and orangutan both exhibit the same nucleotide at the orthologous position. Of the 27,538 nonsynonymous substitutions, a total of 21,278 are fixed in all African, Asian and European populations. Of the 40,709 synonymous substitutions, 32,666 are fixed. The ratio of the number of fixed nonsynonymous to fixed synonymous substitutions is 65.1 %, which is in very good agreement with the previous result of 64% obtained by Boyko *et al.* (Boyko et al. 2008). Only diversity patterns close to fixed substitutions are analyzed in the near versus far and the functional versus non-functional tests. Focusing on fixed substitutions is therefore intended to make results easier to interpret. This is also expected to be conservative when searching for sweeps, because we exclude fixations that occurred after the split of African and non-African populations.

PolyPhen-2 analysis

We use PolyPhen-2 (Adzhubei et al. 2010) to identify which human-specific amino acid substitutions are more likely to be functionally consequential. PolyPhen-2 annotates SNPs but can also be used to annotate fixed amino acid changes by using the REVERSE option. Of the 21,278 fixed amino acid changes specific to the human lineage, 18,924 (89%) can be annotated. Of these, 15,488 are annotated as benign, 1,874 as possibly damaging, and 1,562 as probably damaging by PolyPhen-2. The possibly damaging and probably damaging amino acid changes (18% of the total) are more likely to be functionally consequential than the benign ones. Thus, in the functional versus non-functional test (Main text and Bootstrap procedure below), functional windows are those close to a possibly or probably damaging amino acid change and the

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non-functional windows are those close to a benign amino acid change, but far from any possibly or probably damaging one.

Neutral diversity

Neutral diversity is measured using average heterozygosity π, measured as $2f(1-f)n/(n-1)$ where f is the frequency of the non-reference allele in the 1000 Genomes phase 1 20100804 release (December 2010 update) and n is the number of chromosomes in 500 kb windows (see below for an in-depth discussion on window size). More specifically, average heterozygosity is calculated separately for the three African, Asian and European populations. We use only positions outside of CDS, UTRs (from Ensembl v64) and phastCons CNEs (from the UCSC Genome Browser), simple repeats and transposable elements identified by Repeatmasker (http://genome.ucsc.edu/). Excluding functional elements, repeats and positions not aligned with a nucleotide in macaque, approximately a third of the positions within windows can be used on average to measure neutral diversity. We also exclude all windows closer than 5 Mb to centromeres or telomeres from our analysis. Diversity is further scaled by the number of positions found to be divergent between human and macaque in human-macaque BLASTZ (Schwartz et al. 2003) alignments retrieved from the UCSC Genome Browser (http://genome.ucsc.edu/). This is done to eliminate the effect of local variations in mutation rate or remaining strong selective constraint. Because local changes in mutation rate and strong selective constraint affect both diversity and divergence equally, using the ratio of diversity on divergence removes at least partially the effects of heterogeneous mutation rates and selective constraint. Using scaled diversity implies that only those positions where a nucleotide (non-N or any other undefined position) is aligned with a nucleotide in macaque are used. Scaled neutral diversity is calculated within 500 kb windows sliding every 5 kb in the genome. See Supplemental Material for a complete explanation on why we choose large windows with a fixed physical rather than genetic size.

Bootstrap procedure

In humans local functional density is very heterogeneous and is a main determinant of neutral diversity. Regions of high functional density have higher levels of BGS and hence lower levels of neutral diversity (McVicker et al. 2009; Lohmueller et al. 2011). GC content and recombination also have a strong influence on levels of neutral

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649 diversity (Results). In our study we want to characterize the effect of positive selection
650 on neutral diversity. This is done by comparing neutral diversity in regions of the genome
651 where the rate of positive selection is expected to be higher with neutral diversity in
652 regions where the rate of positive selection is expected to be lower. Genomic windows
653 with potentially higher rates of positive selection are called tested windows, and genomic
654 windows with potentially lower rates of positive selection are called control windows. In
655 the near versus far test, tested windows are the windows near amino-acid changes
656 (nearest amino acid change at less than 0.1 cM from the center of the window) and the
657 control windows are windows far from any amino-acid change (>0.5 cM). In the
658 functional versus non-functional test, tested windows are the windows near functional
659 amino acid changes according to PolyPhen-2 (<0.1 cM) and the control windows are
660 windows near non-functional amino-acid changes (<0.1 cM) but far from any functional
661 amino-acid change (>0.5 cM). In addition to positive selection, we also tested whether
662 windows very far from any amino acid change (>1 cM) experience more BGS than
663 windows moderately far from amino acid changes (between 0.1 cM and 1 cM). In this
664 case tested windows are the windows between 0.1 cM and 1 cM and control windows
665 are the windows further than 1 cM from any amino acid change.
666 The major challenge when testing positive selection by comparing tested and
667 control windows is to make sure that both kinds of windows are as similar as possible.
668 One may think of an example where in tested windows the percentage of positions
669 within CDS is 2% on average and only 0.5% in control windows. In this case there are
670 four times more CDS in the tested windows than in the control windows. BGS is thus
671 stronger in the tested windows. In such an example, neutral diversity is lower in tested
672 windows than in control windows not because of positive selection but because of
673 stronger BGS, and it is impossible to conclude anything about positive selection. This
674 example shows that in order to be conclusive about positive selection we need to
675 compare windows with levels of BGS as similar as possible. This means that the tested
676 and control windows need to have on average similar functional densities, in addition to
677 similar recombination rates and GC content. This is achieved by using a simple
678 bootstrap procedure. For each tested window, we match a control window whose
679 characteristics are not more different than fixed thresholds compared to the tested
680 window. These characteristics are the average recombination rate in the window
681 obtained from the most recent deCode 2010 genetic map (Kong et al. 2010), GC
682 content, CDS density (Ensembl v64), conserved coding sequences (CCDS) density

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683 (Ensembl v64), UTR density (Ensembl v64), and total functional density (TFD). CCDS
684 are the 83% of coding sequences that overlap conserved segments (mammal-wide
685 and/or primate-wide) predicted by phastCons (Siepel et al. 2005) and available at the
686 UCSC Genome Browser (phastCons applied to a genome alignment of 44 mammals).
687 TFD is the percentage of positions in a window that are in at least one of these different
688 types of functional elements: CDS, CCDS, UTR, phastCons conserved non-coding
689 element (CNE). In addition, we also control for the amount of surrounding CDS, which is
690 the number of positions within a CDS up to 0.1 cM upstream and 0.1 cM downstream of
691 a window.
692 For each tested window, we find a matching control window whose
693 recombination, GC content, CDS, CCDS, UTR, TFD and surrounding CDS are
694 comprised between x% and y% of their values in the tested window. The values of x and
695 y are specific to each of the controlled factors, and x is smaller than one while y is
696 greater than one. For example we could ask control windows to have a CDS density
697 comprised between x=80% and y=120% of the tested window CDS density. In practice
698 we adjust the thresholds so that when the bootstrap is complete tested windows and
699 control windows have a very similar average recombination rate, average GC content,
700 average CDS, CCDS, UTR, TFD and surrounding CDS. In addition, we also make sure
701 that they have very similar phastCons CNE density.
702 Although we cannot avoid slight differences, we make sure they are in the
703 conservative direction. For example the average CDS density in the control windows
704 may be 3% higher than in the tested windows, and the average recombination rate may
705 be 5% lower. When no matching control window is found in the genome the tested
706 window is excluded from the analysis. The same control window can be used several
707 times as a match for several tested windows. The different amounts of sequences that
708 could be used for each test are shown in Supplemental Table 1. The x% and y%
709 thresholds used for the different tests conducted in this analysis are provided in
710 Supplemental Table 3. Note that the thresholds were adjusted so that they could be
711 used for all the repetitions of a given test in various conditions. For example in the near
712 versus far test we used thresholds that can be applied whether or not we use only
713 windows below a fixed recombination threshold, and whether or not we use only low
714 CCDS windows (Results). This is to ensure that the results obtained under these
715 different conditions can be fairly compared between each other. For the near vs. far test,
716 low CCDS and recombination rate lower than 1 cM/Mb (Results), we further controlled

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717 that the observed decrease of diversity is robust to changing x and y for several factors,
718 while still having conservative comparisons between near and far windows
719 (Supplemental Table 4).
720 For each test the bootstrap procedure is conducted ten independent times. Each time
721 we calculate the average neutral diversity in tested windows Π_{tested}, the average neutral
722 diversity in control windows $\Pi_{control}$ and the ratio $\Pi_{tested}/\Pi_{control}$. Different realizations of the
723 bootstrap procedure give very similar $\Pi_{tested}/\Pi_{control}$ ratios. For all tests and for each
724 realization the ratio $\Pi_{tested}/\Pi_{control}$ never differs by more than 10% of its average over the
725 ten realizations. The observed ratios $\Pi_{tested}/\Pi_{control}$ shown in Figs. 1 and 2 represent the
726 average over the ten realizations of the bootstrap procedure. Because there is so little
727 variation between the different realizations of the bootstrap procedure, we always use
728 the first realization for running populations simulations (see Population simulations
729 below) and for calculating *P*-values of the randomization test (see Randomization test
730 below). Note also that we do not include average sequencing depth in the windows as
731 one of the controlled variables although it is well known to have an effect on the
732 estimation of neutral diversity. This is because we found this is not necessary since on
733 average the tested and control windows retained by the bootstrap procedure have
734 extremely similar average sequencing depths that never vary by more than 0.5% from
735 each other.
736
737 **Randomization test**
738 We use a randomization test to estimate the significance of the differences of neutral
739 diversity we observe between tested and control windows used in the bootstrap
740 procedure. In order to obtain a random distribution of $\Pi_{tested}/\Pi_{control}$ for a given realization
741 of the bootstrap procedure, we need to shuffle tested and control windows while
742 accounting for a number of features of the analysis. First, the tested and the control
743 windows are often clustered together, much like the windows represented along a
744 chromosome in Supplemental Fig. 5. Π_{tested} and $\Pi_{control}$ are calculated from groups of
745 neighboring, overlapping windows that have correlated neutral diversity values.
746 Compared to a situation where we would have the same number of windows but all
747 independent from each other, this grouping substantially increases the variance of
748 Π_{tested}, $\Pi_{control}$ and thus of the ratio $\Pi_{tested}/\Pi_{control}$. Shuffling individual windows
749 independently from each other is therefore very likely to greatly underestimate the true

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variance of the ratio. Second, during the bootstrap procedure the same control window can be matched with several tested windows, which should also be taken into account during the randomization process. In order to maintain the structure of the sampling scheme used in the bootstrap procedure, we shuffle blocks of neighboring windows (Supplemental Fig. 4). Windows used in the bootstrap procedure are first ordered according to their genomic positions. We then cut 20 segments of equal size (Supplemental Fig. 4 represents a situation with only three segments). This is done to maintain the grouping of windows. The 20 segments are then shuffled to obtain a new random ordering of windows. In addition a segment can be flipped with a probability of 50%. The same sampling scheme that was used during the bootstrap procedure is finally applied to the randomized windows. For example in the genome the positions 19,20 and 21 are occupied by tested windows tested_19, tested_20 and _tested_21 that are all matched with the same control window control_29 at position 29 (Supplemental Fig. 2). After the randomization, positions 19, 20 and 21 are now occupied by tested windows tested_8, tested_9 and tested_10 that now all match with window tested_18 at position 29. This way the neighboring windows tested_19, tested_20 and tested_21 have been replaced by three other neighboring windows, and window tested_18 matches three times as window control_29. The randomization process is repeated 10,000 times to obtain the *P*-value for the test. *P*-values are calculated as the proportion of randomizations where random $\Pi_{tested}/\Pi_{control}$ is lower or higher than the observed $\Pi_{tested}/\Pi_{control}$ depending on the case studied. This means that the randomization test is a one-sided test.

Population simulations

In our study we use forward simulations to estimate the ranges of the ratios of Π_{near}/Π_{far} and $\Pi_{func}/\Pi_{non\text{-}func}$ under both a demographic scenario of panmixia with no advantageous mutation and under a scenario of panmixia with different rates and strengths of positive selection. Simulations were conducted using SLiM (Messer 2013). We simulate segments of the human genome where windows were sampled by the bootstrap procedure. Supplemental Fig. 3 shows how those segments are defined based on where the sampled windows are in the genome and how far they are from each other. In Supplemental Fig. 5, 500 kb sampled windows define three non-overlapping groups along a chromosome. The first and second groups (starting from the left) are at distance of 0.23 cM from each other. These two groups are fused together to form a genomic

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784 segment that includes them both. The segment is further extended 0.1 cM upstream and
785 0.1 cM dowstream to avoid edge effects and to include the effect of eventual neighboring
786 advantageous mutations not included in, but close to, the sampled windows
787 (Supplemental Fig. 5). The third group is at 0.84 cM and is treated as an independent
788 segment. Overall, groups of windows closer than 0.5 cM from each other are fused
789 together while groups further than 0.5 cM from each other are treated as independent
790 simulated segments.
791 All the segments in the genome are simulated independently and the simulated ratios
792 $\Pi_{tested}/\Pi_{control}$ are calculated exactly as they are using the bootstrapping procedure. This
793 means that the same 500 kb windows are used and that within each window, variants
794 whose coordinates fall within a functional element, a repeated element, or do not align
795 with macaque in the real genome, are excluded from the calculation of simulated
796 diversity. The whole operation is repeated 100 times for the estimation of confidence
797 intervals of $\Pi_{tested}/\Pi_{control}$.
798 The recombination maps used in each segment match the deCode 2010
799 recombination map (Kong et al. 2010). The simulations were conducted using a
800 population of 500 individuals and the recombination and mutation rates were rescaled
801 accordingly to match the average recombination rate (1.16 cM/Mb) and the average
802 heterozygosity (0.001) observed in the human genome. After a burn-in of 5,000
803 generations, the neutral simulations are continued for 1000 additional generations (this
804 is equivalent to 20,000 generations in a non-rescaled 10,000 individuals human
805 population). Simulations with positive selection are continued for 2,500 generations after
806 the burn-in to ensure that all advantageous mutations introduced after the burn-in are
807 given a fair amount of time to fix.
808 For the simulations with positive selection, we introduce advantageous mutations
809 at random generation times with a fixed rescaled selection coefficient at positions where
810 amino acid changes are found in the human genome. As an example, we can simulate a
811 scenario where 10% of the amino acid changes were adaptive with s=1%. The selection
812 coefficient of 1% in a 10,000 individuals population is rescaled to 20% in our 500
813 individuals simulated population to maintain the same intensity of selection. In order to
814 obtain 10% of fixed adaptive mutations, given the probability of fixation ($2s$=40%) we
815 need 25% of the introduced mutations with s=20%. These advantageous mutations are
816 introduced randomly among all the locations with an amino acid change. For the sake of
817 speed in our simulations with positive selection we use 2,500 generations after burn-in

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although in our rescaled population the number of generations to the human-chimpanzee most recent ancestor is 10,000 generations (rescaled from 200,000 generations assuming a TMRCA of 5 My and a generation time of 25 years). Advantageous mutations were thus attributed an introduction time between 1 and 10,000 generations after burn-in, but only those mutations having a random introduction generation between 1 and 2500 were actually introduced in the population.

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Figures legends:

Figure 1. Lower diversity near versus far from amino acid substitutions
Each panel shows the level of synonynmous heterozygosity near amino acid changes (π_{near}) compared with that far from amino acid changes (π_{far}) for the particular subpopulation. π_{near} is the average over all near windows (<0.1 cM) from the bootstrap procedure (Methods). π_{far} is the average over all far windows (>0.5 cM but <1 cM). The grey area depicts the 95% confidence intervals based on neutral simulations (Methods). The dashed lines show 95% and 97.5% confidence intervals based on a randomization test (Methods). Randomization tests result in at most 35% greater variance than the neutral simulations. This is expected given that neutral simulations do not account for complex demography and other sources of noise in the data. On the x axis, ≤1, ≤1.5 and so on means that we use only windows with recombination rates lower or equal to 1 cM/Mb, 1.5 cM/Mb and so on to compare diversity near and far from amino acid substitutions. “all” means that we use all windows independently of their recombination rates.

Figure 2. Lower diversity near functional amino acid substitutions
We compared heterozygosity near functional amino acid changes π_{func} with heterozygosity near non-functional amino acid changes $\pi_{non\text{-}func}$. π_{func} is the average over all functional windows from the bootstrap procedure (Methods). $\pi_{non\text{-}func}$ is the average over all non-functional windows. The grey area depicts the 95% confidence intervals based on the neutral simulations (Methods). The dashed lines show 95% and 97.5% confidence intervals established based on a randomization test (Methods). On the x axis, ≤1, ≤1.5 and so on means that we use only windows with recombination rates lower or equal to 1 cM/Mb, 1.5 cM/Mb and so on to compare diversity near and far from functional amino acid substitutions. “all” means that we use all windows independently of their recombination rates.

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Figure 3. Combined near versus far and functional versus non-functional tests
Clouds of small dots represent the ratios π_{near}/π_{far} and $\pi_{func}/\pi_{non\text{-}func}$ obtained with the randomization test. The larger dot in each graph represents the observed π_{near}/π_{far} and $\pi_{func}/\pi_{non\text{-}func}$. The numerical values at the lower right side of each graph are the P-values obtained after 10,000 iterations of the randomization test. The P-values are estimated as the proportion of the randomizations that give values below the observed value in both tests.

Figure 4. Robustness of *iHS* and *XPEHH* to BGS

We tested the effect of BGS on *iHS* and *XPEHH* (Results and Supplemental Material). Upper row: average heterozygosity. Middle row: *iHS*. Lower row: *XPEHH*. The full lines represent average *iHS* or *XPEHH* along the simulated region. The dashed lines represent the limits of *iHS* or *XPEHH* 95% confidence intervals.

Figure 5. Stronger haplotype and SFS based signals of positive selection near amino acid changes.
This figure shows the comparison of the top 10%, 5%, 2% and 1% *XPEHH* and *iHS* windows near and far from amino acid changes. * randomization test $P \leq 0.05$; ** $P \leq 0.01$; *** $P \leq 0.001$. Left side of histograms: all regions irrespective of recombination rates. Rigth side: only regions with recombination rates lower than 0.5 cM/Mb. The max(EA,EU) histograms show the results obtained when retaining for each window the maximum signal of the East Asian and European populations.

Figure 6. Simulated decreases of diversity for different rates and strengths of positive selection
We ran 100 forward simulations (Methods) to estimate the average and 95% confidence intervals (CI) for the decrease of diversity near amino acid changes under different rates and strengths of positive selection. To be conservative, we extended the confidence intervals from simulations by 35% given that neutral simulations underestimate the variance by ~35% in the simulated regions as shown in Fig. 1.

Figure 7. Most human recent positive selection occurs in regulatory sequences

The filled circles and squares show the correlation coefficients of the absolute values of *iHS* with the density of regulatory and coding sequence density, respectively, controlling for recombination and average pairwise diversity (Methods). The open circles and squares show partial correlations. For instance, an open circle shows partial correlation between absolute values of *iHS* and regulatory density controlling for coding density (also controlling for recombination and average pairwise diversity). All correlations are

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Spearman's rank correlations or partial correlations. Correlation coefficients greater than 0.05 are all highly significant ($P<2x10^{-16}$).

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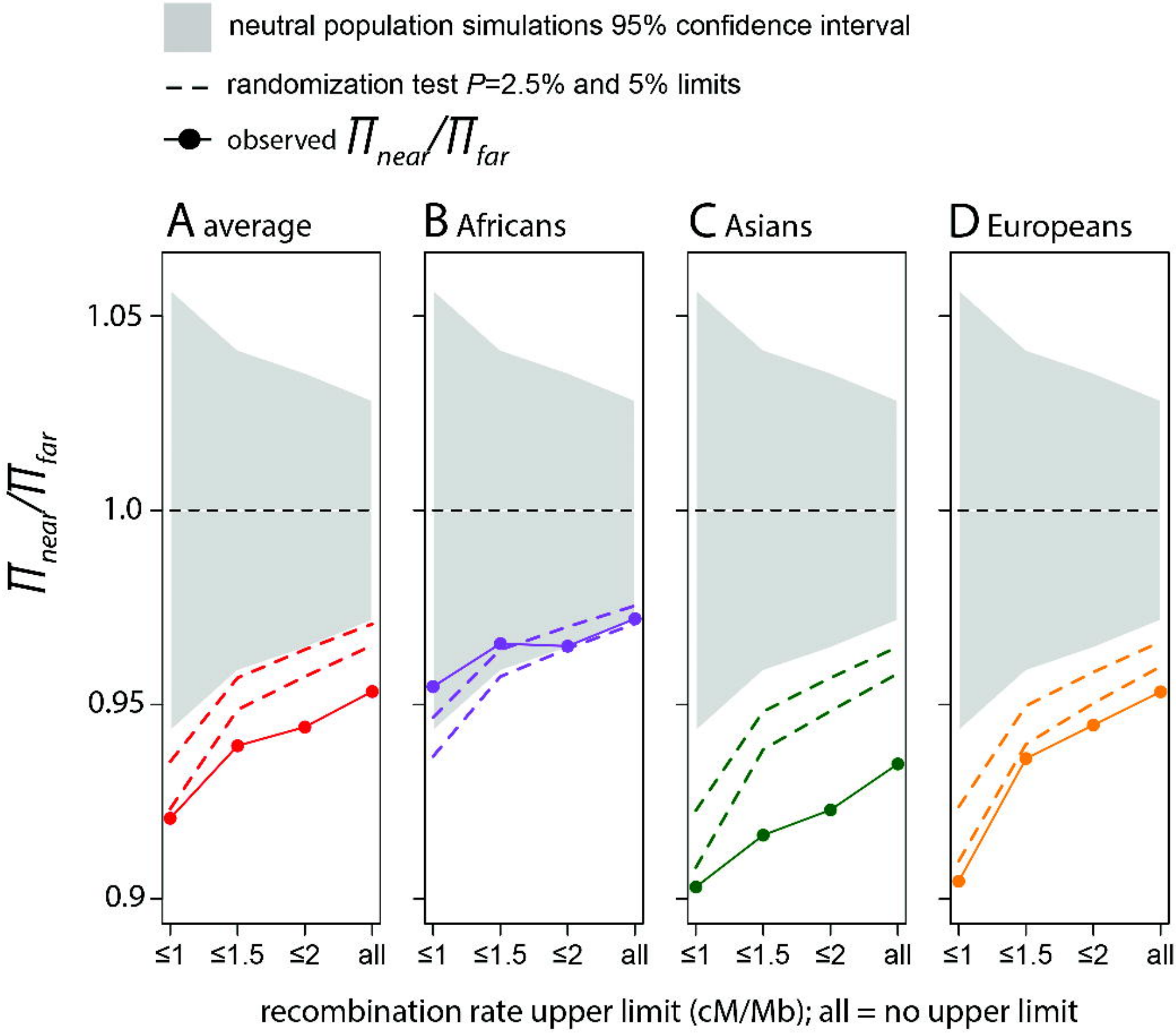
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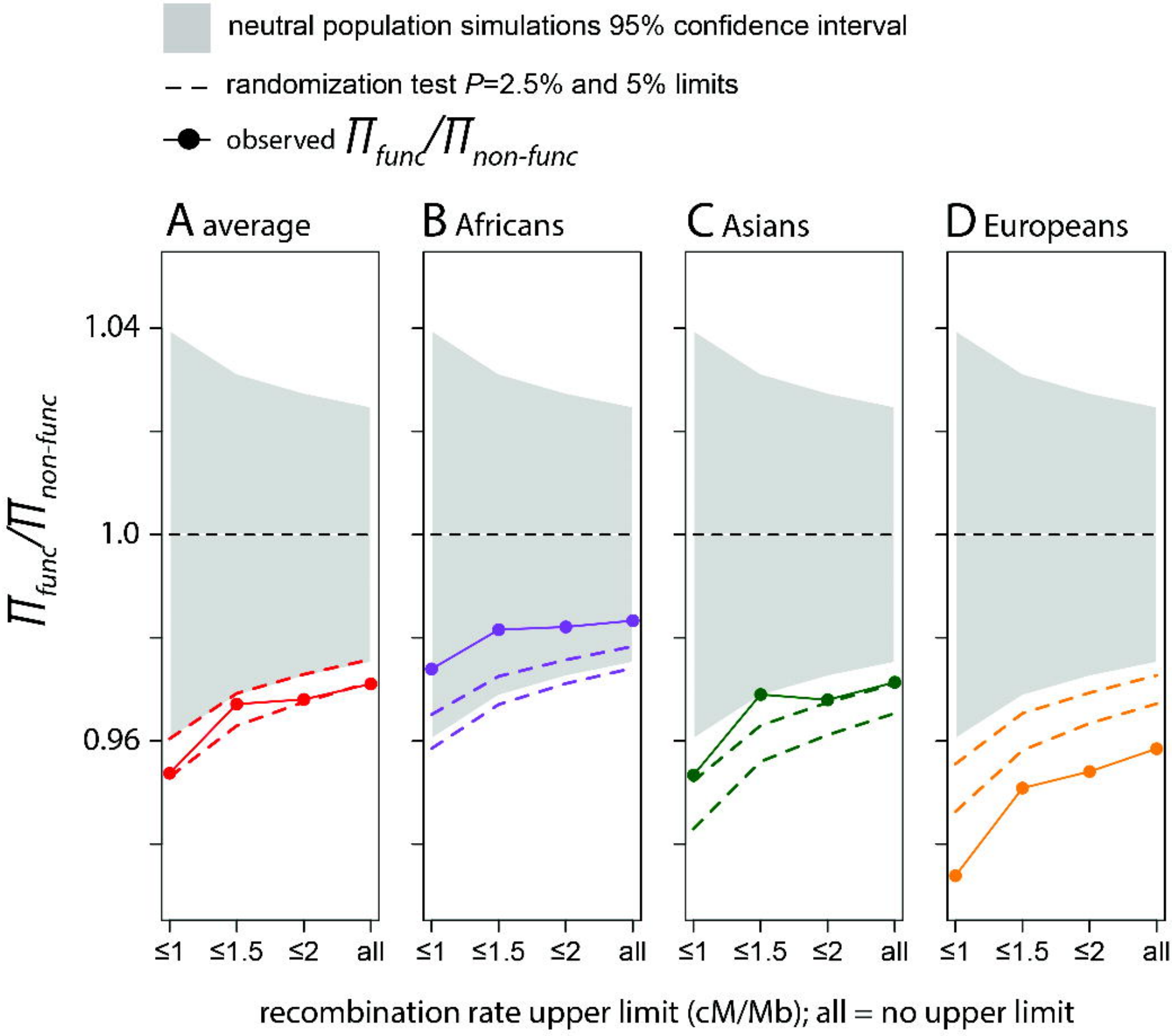
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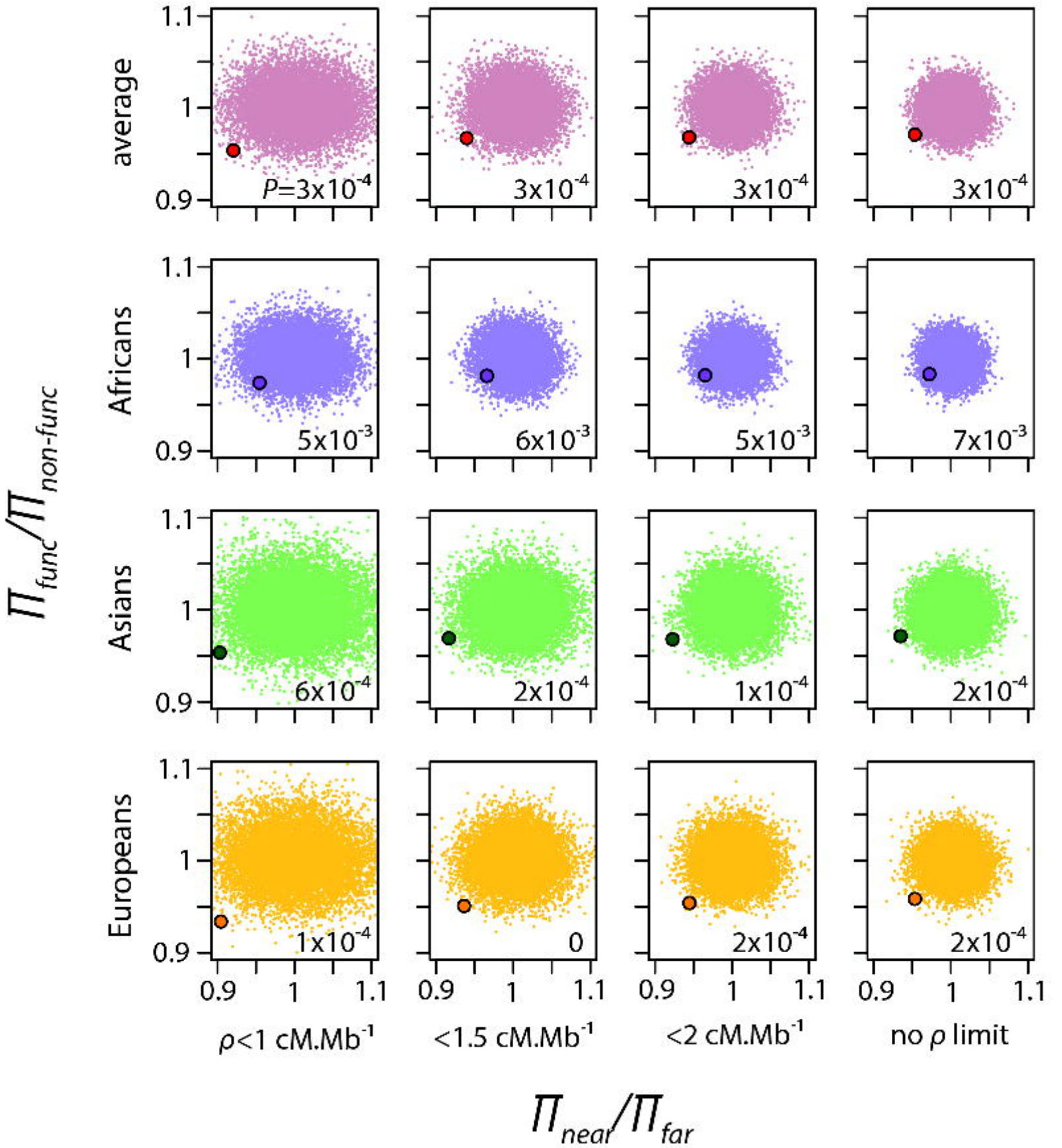
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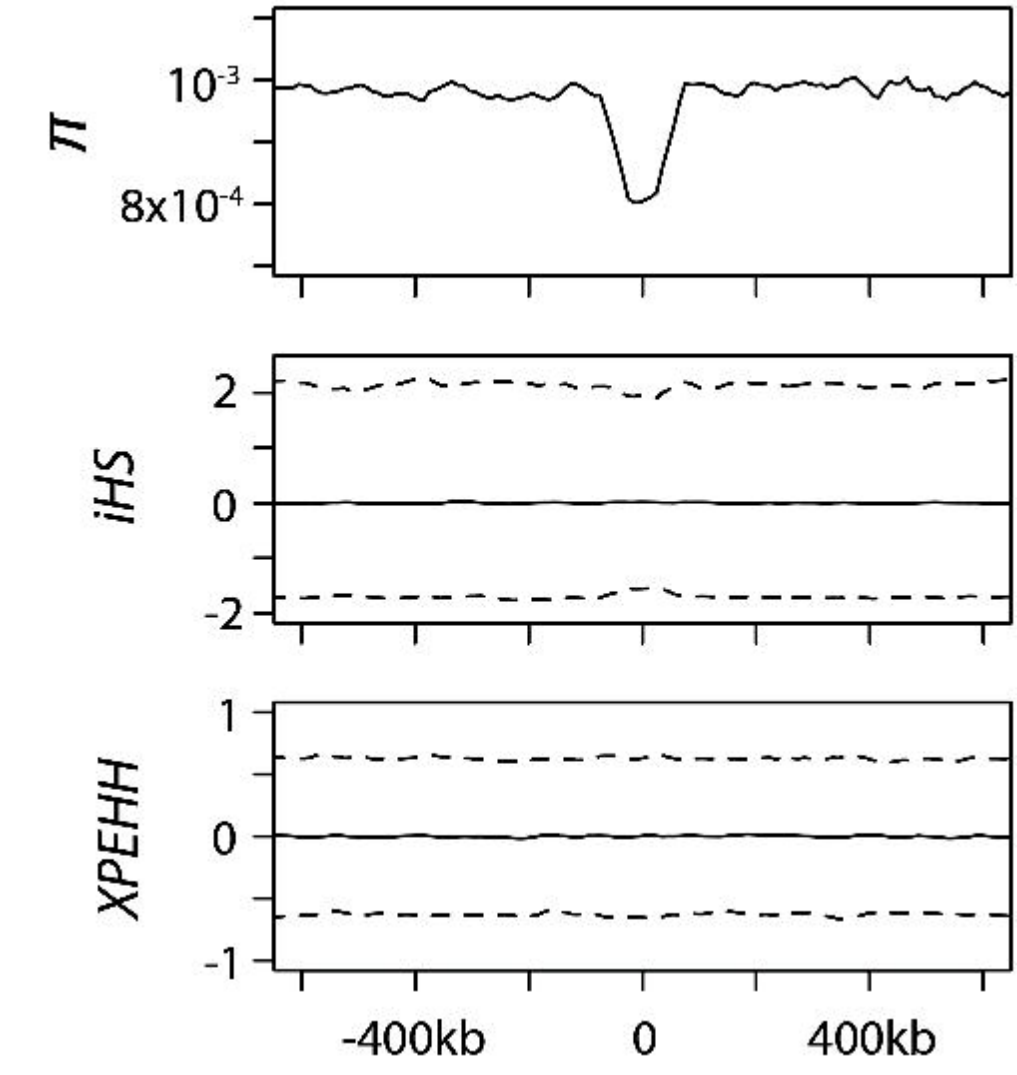
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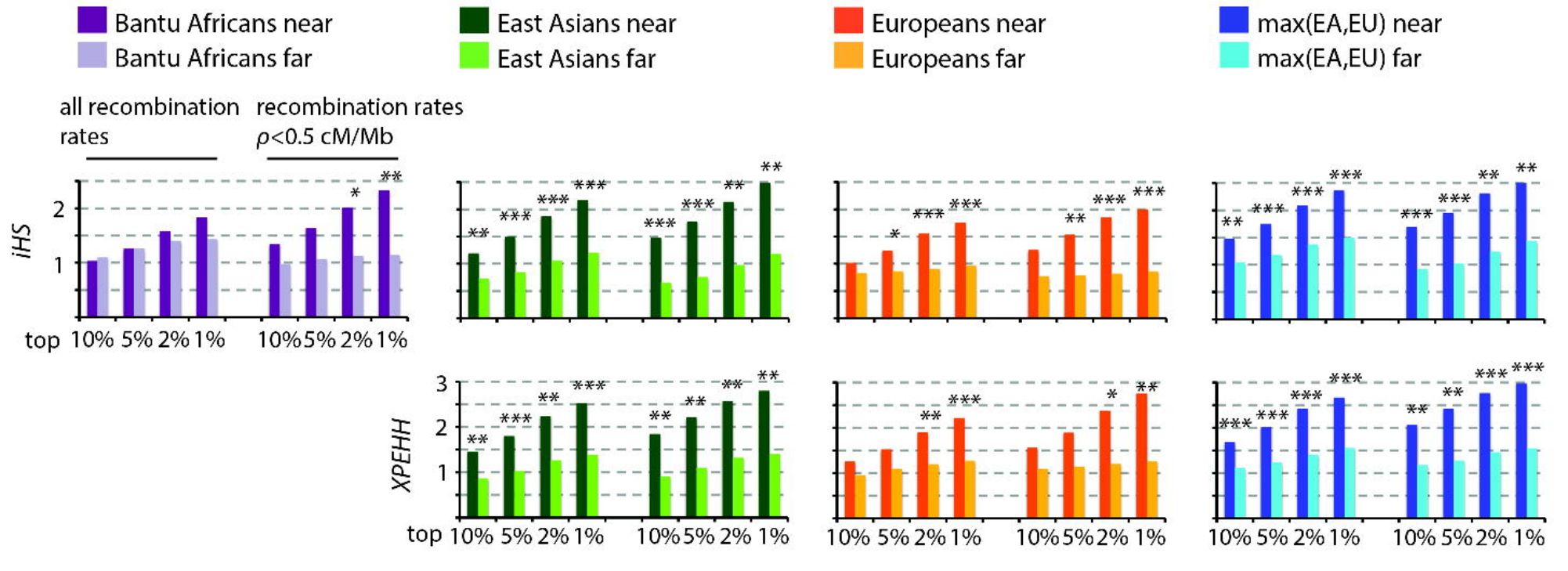
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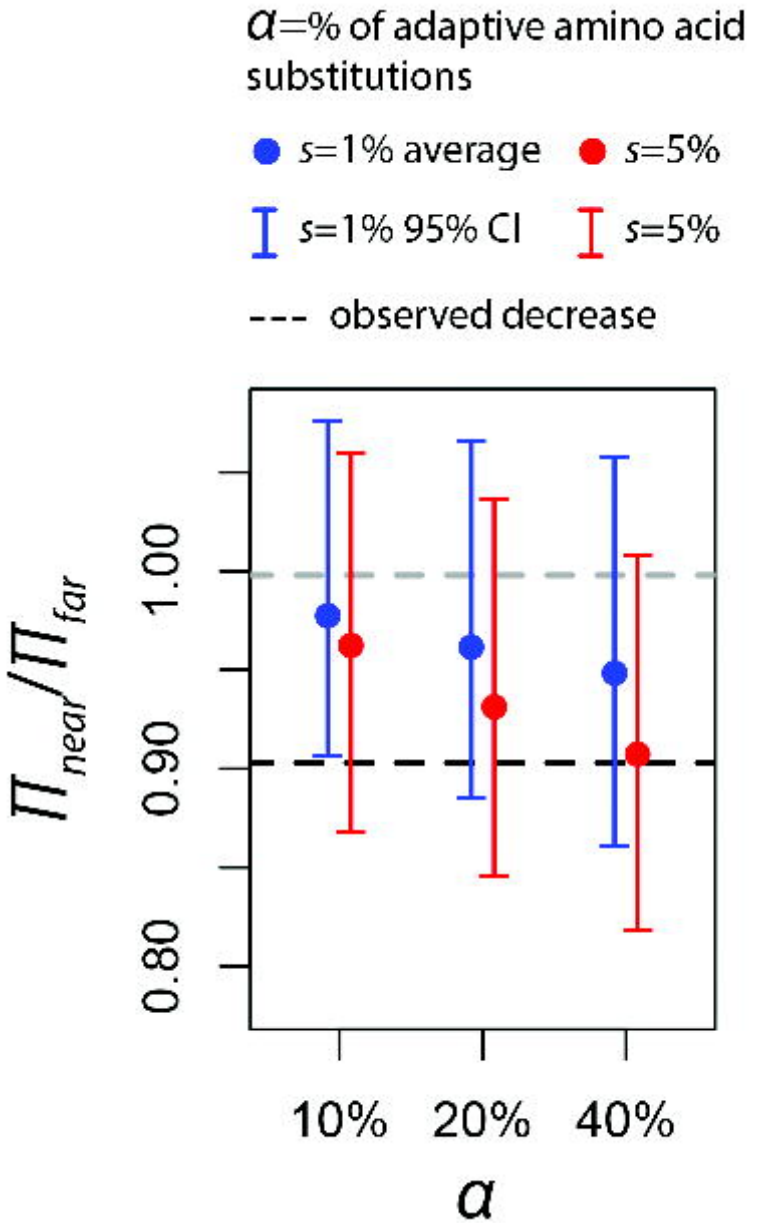

neutral population simulations 95% confidence interval
randomization test P=2.5% and 5% limits
observed π_{near}/π_{far}
A average
B Africans
C Asians
D Europeans
π_{near}/π_{far}
1.05
1.0
0.95
0.9
≤1
≤1.5
≤2
all
recombination rate upper limit (cM/Mb); all = no upper limit

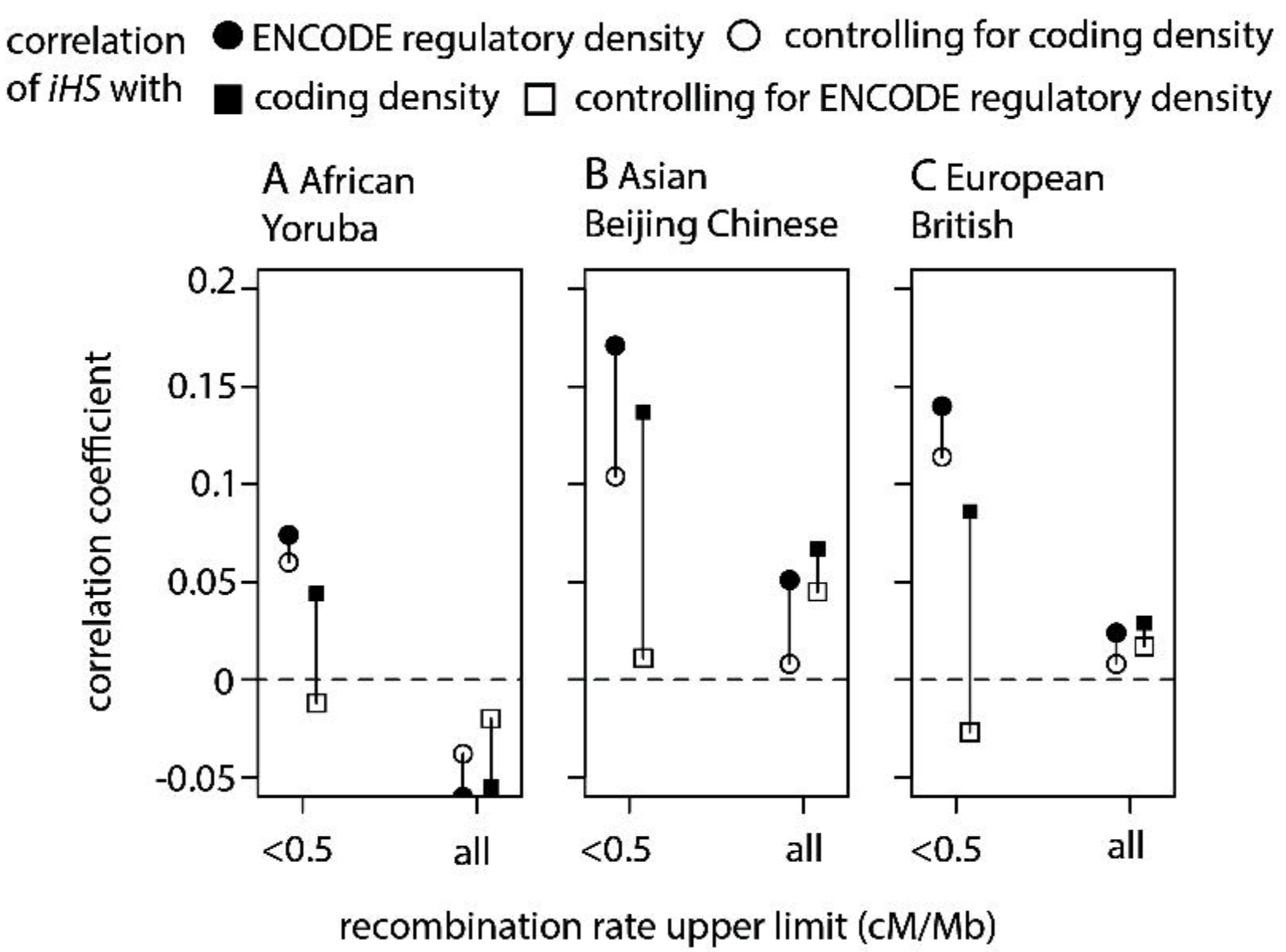

neutral population simulations 95% confidence interval
randomization test P=2.5% and 5% limits
observed $\pi_{func}/\pi_{non\text{-}func}$
A average
B Africans
C Asians
D Europeans
1.04
1.0
0.96
$\pi_{func}/\pi_{non\text{-}func}$
≤1
≤1.5
≤2
all
recombination rate upper limit (cM/Mb); all = no upper limit


average
Africans
Asians
Europeans
P=3x10-4
3x10-4
3x10-4
3x10-4
5x10-3
6x10-3
5x10-3
7x10-3
6x10-4
2x10-4
1x10-4
2x10-4
1x10-4
0
2x10-4
2x10-4
1.1
1
0.9
0.9
1
1.1
ρ<1 cM.Mb-1
<1.5 cM.Mb-1
<2 cM.Mb-1
no ρ limit
Π func/Π non-func
Π near/Π far


π
10-3
8x10-4
iHS
2
0
-2
XPEHH
1
0
-1
-400kb
0
400kb


Bantu Africans near
Bantu Africans far
East Asians near
East Asians far
Europeans near
Europeans far
max(EA,EU) near
max(EA,EU) far
all recombination rates
recombination rates ρ<0.5 cM/Mb
iHS
XPEHH
top 10% 5% 2% 1% 10% 5% 2% 1%


α=% of adaptive amino acid substitutions
s=1% average
s=5%
s=1% 95% CI
s=5%
--- observed decrease
Π near/Π far
1.00
0.90
0.80
10%
20%
40%
α


correlation of *iHS* with
ENCODE regulatory density
controlling for coding density
coding density
controlling for ENCODE regulatory density
A African Yoruba
B Asian Beijing Chinese
C European British
correlation coefficient
0.2
0.15
0.1
0.05
0
-0.05
<0.5
all
recombination rate upper limit (cM/Mb)